



## An HIV USB Stick?

There's a fast and highly accurate USB stick that uses a drop of blood to test for the virus in less than 30 minutes. <http://dgit.in/USB-hiv>



## The Rubik's Cube Robot

Infineon built a robot called Sub1 Reloaded that solve the Rubik's Cube in 637 milliseconds. <http://dgit.in/robosub1>

Origins

# CLICK!

PHOTOGRAPHY'S JOURNEY  
OVER HISTORY TOWARDS THE  
PERFECT SHOT!

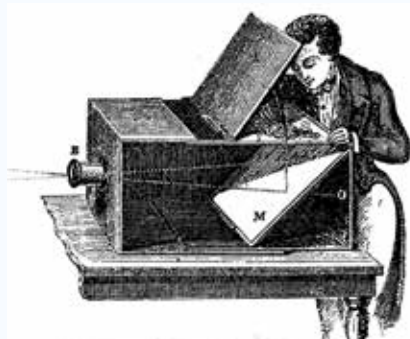
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Pictures are everything today. Used to be a time they were reserved for special occasions, but now, we click them all the time. This is because taking a picture is as easy as blinking an eye now – literally, there are blink detecting selfie cameras on phones! If you're old enough to remember the film camera days, you will know that taking pictures wasn't always this easy. Interestingly, film photography wasn't the oldest technology, because it all started in the 1800s.

## How it all began

If you have played with a pinhole camera as a kid, the first cameras were pretty much just that. This is a natural phenomenon where you can see things on the other side of a pinhole in a screen, but the image is reversed and inverted. This is also how the human eye sees.

Recording light (taking a picture) came about in 1816, when Nicephore Niepce put silver chloride to use by coating paper with it and subjecting it to the camera obscura effect. The results weren't really permanent and Niepce turned his attention towards other substances. The coating was changed to bitumen, a naturally occurring petroleum tar and the process was refined by Niepce and Louis Daguerre in using a better resin and a different post processing technique that yielded better



The basic principle behind cameras - camera obscura

results. But exposure times for capturing images were still measured in hours.

Exposure times didn't come down until Daguerre made the discovery that slightly invisible images made on photographic plates due to shorter exposure times could be developed later into fully visible images by exposure to mercury fumes. This single discovery brought exposure time down from hours to a few minutes, and in 1839 it was made publicly available.

After a series of refinements and experiments that gradually improved the process of plate-based photography, something happened in 1884 that changed it forever. George Eastman of New York developed dry gel on paper, i.e. film, to replace the voluminous photographic plates and dangerous toxic chemicals that photographers had to carry everywhere. In 1888